

ISHWARYA ANANDAKRISHNAN

Raleigh, NC (Open to Relocation) | 984-379-2052 | ishwaryaanandakrishnan@gmail.com
[linkedin.com/in/ishwarya-anand](https://www.linkedin.com/in/ishwarya-anand) | github.com/ishwarya0103

SUMMARY

MS Computer Science (May 2026) with 2+ years in **ML infrastructure**, **model evaluation**, and **generative AI** — spanning model deployment, **data processing**, optimization, and **reinforcement learning**. Experience implementing **GenAI** features in production, designing **experimentation frameworks**, and applying **natural language processing** at scale in distributed systems.

SKILLS

Languages: Python, Java, TypeScript, SQL, C++, Bash
ML & Modeling: PyTorch, TensorFlow, Scikit-learn, CNNs, Transformers, reinforcement learning, conformal prediction, MC Dropout, Flash Attention, spaCy, XGBoost
GenAI & LLMs: LangChain, LangGraph, LlamaIndex, OpenAI APIs, RAG, PEFT/LoRA/QLoRA, Hugging Face, RA-GAS, FAISS, Qdrant, MongoDB Atlas
Infrastructure: Docker, Kubernetes, AWS, GCP, MLflow, Airflow, Prometheus, Grafana, CI/CD, PostgreSQL, Git

EXPERIENCE

Research Assistant, Generative AI & ML Infrastructure, NC State University Jan 2025 – Present
Raleigh, NC

- Developed a production **GenAI** Q&A system (RAG, LangChain, LangGraph, OpenAI GPT-4, MongoDB Atlas) with multi-step reasoning, and REST API interfaces; serves **50+ researchers** over **1,000+ documents** at **sub-3s** latency
- Designed a **model evaluation** and **experimentation** framework (RAGAS: faithfulness, context recall, hallucination); systematic **data processing** experiments across chunking, embedding, and reranking strategies improved **precision 12%**
- Implemented **ML infrastructure** on AWS ECS; containerized inference with Docker, Kubernetes **HPA autoscaling**, and Prometheus/Grafana debugging and observability; **CI/CD** reduced deployment cycle **70%**

Software Engineer, AI & Distributed Systems, Strand Life Sciences Jul 2023 – Jul 2024
Bangalore, India

- Engineered distributed REST APIs and ETL pipelines (Python, Java, AWS) processing **8B+** genomic records; reduced execution time **30%** through parallelized containerized microservices and large-scale **data processing** optimizations
- Developed **NLP** entity-extraction models (spaCy, Hugging Face) on large-scale clinical text; evaluated with F1, precision-recall, and AUC across **3 production** ML services; documented **model optimization** methodology for team reuse
- Automated reporting workflows via Python and Google APIs, eliminating **4+ hours/day** of manual processing and unblocking **model retraining** across cross-functional teams

Undergraduate Researcher, ML Algorithms, NIT Trichy Aug 2021 – May 2023
Trichy, India

- Implemented **reinforcement learning** agents (Deep Q-Network) for **sequential decision making** in dynamic network optimization, improving throughput **15%**; proposed the Fitness Dependent Optimizer, awarded **Best Presentation** among **45 teams** at Pragyan Tech Fest; presented at **IC@MACE 2023**

Software Engineering Intern, Strand Life Sciences May 2022 – Jul 2022
Bangalore, India

- Delivered SSO server (CAS + Google OAuth) securing **7+** internal tools; wrote automated test suites improving testability and efficiency; deployed ELK Stack debugging and observability, reducing incident MTTR by **20%**

PROJECTS

Model Evaluation: Uncertainty Quantification Framework github Jan 2025
PyTorch, Scikit-learn, Conformal Prediction, MC Dropout, NumPy

- Built a **model evaluation** and calibration framework using conformal prediction and MC Dropout across regression and classification tasks; achieved **90–95%** empirical coverage; benchmarked ECE, calibration curves, and reliability diagrams over **5+** architectures

Scalable ML Infrastructure: Kubernetes Deployment and CI/CD github Nov 2024
FastAPI, Kubernetes, Docker, MLflow, Airflow, Prometheus, Grafana, AWS

- Designed and deployed a production **ML serving** layer on Kubernetes with **HPA autoscaling** and Prometheus/Grafana monitoring; orchestrated **10+** experiment runs via MLflow and Airflow; **CI/CD** cut redeployment steps **80%**

Clinical NLP Transformer: Flash Attention and Inference Optimization github Oct 2024
PyTorch, Flash Attention, Hugging Face Transformers, FastAPI

- Implemented a decoder-style transformer with Flash Attention achieving **2×** throughput over standard attention; built **model evaluation** harness for NLP classification achieving **0.75 F1** on Portuguese medical text

EDUCATION

M.S., Computer Science, NC State University, Raleigh, NC GPA: 3.7/4.0 | Aug 2024 – May 2026
B.Tech., Computer Science & Engineering, NIT Trichy, India GPA: 8.5/10.0 | Jul 2019 – May 2023